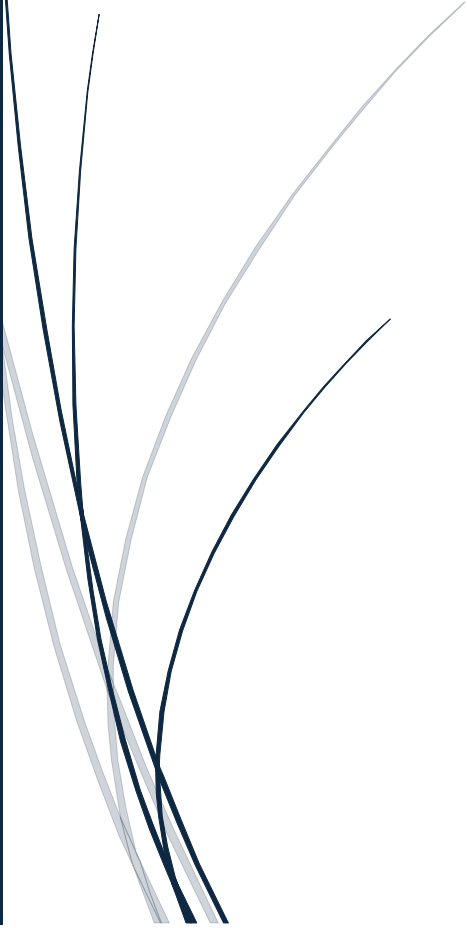


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PROG 6212

PART 01



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PART 01 SECTION B:

An Application Programming Interface (API) is a link/connection between computers or between computer programs.

Below is the API Endpoint design that links the Organiser and Participant interfaces:

1. AUTHENTICATION (Login and Registration)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user account with the appropriate role.	Public	FirstName, Surname, Email, Username, Password, PhoneNumber, Role.	201 Created with account details and confirmation message.
POST	/api/auth/login	Authenticates a registered user and provides authentication details.	Public	Username/Email, Password	200 OK with authenticated user details and authentication token.

2. USER PROFILE

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/profile	Retrieves the profile of the currently authenticated user.	Organiser/Participant	None	200 OK with the authenticated user's profile information.
PUT	/api/profile	Updates the profile information of the currently authenticated user.	Organiser/Participant	Updated profile fields	200 OK with the updated profile information.

3. EVENTS (Viewing, Creating, Updating, and Deleting)

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Retrieves a list of available events.	Retrieves a list of available events.	None	200 OK with a list of events.
GET	/api/events/{id}	Retrieves details of a specific event.	Retrieves details of a	None	200 OK with the selected

			specific event.		events details.
POST	/api/events	Creates a new event.	Organiser	Name, Description, Date, Location, Distance, EventType	201 Created with the newly created event.
PUT	/api/events{id}	Updates an existing event	Organiser	Updated event information	200 OK with the updated event.
DELETE	/api/events{id}	Deletes an existing event.	Organiser	None	200 OK or 204 NO Content confirming deletion

4. CATEGORIES

HTT P Met hod	Route	Descripti on	Role Required	Request Body	Expected Response
GET	/api/events/{eventid e}/categories	Retrieves all categories available for a specific event.	Organiser/ Participant	None	200 OK with a list of categories for the event
GET	/api/categories/{id}	Retrieves details of a specific event category.	Organiser/ Participant	None	200 OK with category details.
POST	/api/events/{eventId} /categories	Creates a new category for an event.	Organiser	CategoryName,Description,MinAge,MaxAge and/or distance information	201 Created with the newly create category.
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	Updated category information	200 OK with the updated category.
DELETE	/api/categories/{id}	Deletes an event category.	Organiser	None	200 OK or 204 Content

5. EVENT ENROLMENT

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments	Enrols a participant into an event by selecting an available category.	Participant	EventID, EventCategoryID	201 Created with the enrolment details and confirmation.
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Participant/Organiser	None	200 OK with enrolment details.
GET	/api/participants/enrolments	Retrieves all events the currently authenticated participant has entered.	Participant	None	200 OK with the participant enrolments
GET	/api/events/{eventId}/enrolments	Retrieves all participants enrolled in a particular event.	Organiser	None	200 OK with a list of enrolments

6. RESULTS

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/results	Captures a participant's finishing result after an event.	Organiser	EnrolmentID, FinishTime, OverallPosition, PositionCategory	201 Created with the record result
GET	/api/results/{id}	Retrieves a specific	Participant/Organiser	None	200 OK with result details,

		participant result.			subject to ownership/access rules.
GET	/api/participants/me/results	Retrieves all results belonging to the authenticated participant.	Participant	None	200 OK with the participant's results/history.
PUT	/api/results/{id}	Updates/corrects an existing result.	Organiser	Updated result information	200 OK with the updated result.
DELETE	/api/results/{id}	Removes an incorrectly captured result.	Organiser	None	200 OK or 204 No Content confirming deletion.

7. ROUTES

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventID}/routes	Retrieves route information for a specific event	Organiser/Participant	None	200 OK with route information
GET	/api/routes/{id}	Retrieves details of a specific route	Organiser/Participant	None	200 OK with route details
POST	/api/events/{eventID}/routes	Adds route information to an event	Organiser	RouteName,Description,RouteStartKm,RouteFinishKm	201 Created with new route

PUT	/api/routes/{id}	Update s existin g route inform ation.	Organiser	Updated route information	200 OK with updat ed route
DELETE	/api/routes/{id}	Delete d route inform ation.	Organiser	None	204 No Conte nt

8. WEATHER

HTT P Met hod	Route	Descr iption	Role Required	Request Body	Expe cted Resp onse
GET	/api/events/{event tld}/weather	Retrie ves stored weath er inform ation for an event	Organiser/ Participant	None	200 OK with weat her infor matio n
GET	/api/events/{event tld}/weather/live	Retrie ves the latest weath er inform ation for the event locati on.	Organiser/ Participant	None	200 OK with curre nt/liv e weat her data
GET	/api/weather/{id}	Retrie ves a specif ic stored weath er record .	Organiser/ Participant	None	200 OK with weat her detail s.
POST	/api/events/{event tld}/weather	Adds weath er inform	Organiser	ForecastDate,ForecastTime,Temper ature,RainProbability,Source,Notes	201 Creat ed with

		atuation to an event.			the weather record
PUT	/api/weather/{id}	Updates stored information.	Organiser	Updated weather information	200 OK with updated weather record
DELETE	/api/weather/{id}	Deletes a stored weather record	Organiser	None	204 No Content

- Below is the API Table Summary that explains how many endpoints each table consists of:

FEATURE	ENDPOINT
Authentication	2
User Profile	2
Events	5
Categories	5
Events Enrolment	4
Results	5
Routes	5
Weather	6
Total RACEDAY API	34 endpoints